

EM616: Time-Series Analysis

Random Processes

Udit Bhatia¹

Arun K. Tangirala²

¹*Department of CE, CSE (IIT Gandhinagar)*

²*Department of Chemical Engineering (IIT Tirupati)*

Introductory remarks

- ▶ **Uncertainty: One of the inescapable truths**
- ▶ **Sources of uncertainty in data-driven modeling:**
 - ▶ Insufficient understanding of processes: *known unknowns* and *unknown unknowns*
 - ▶ Measurement errors
 - ▶ Effect of unmeasured causes

Random and Deterministic Process Modeling

- ▶ **All sources of uncertainties lead to the same repercussion:**
It is not possible to accurately predict a process, giving rise to random processes.
- ▶ **Deterministic processes:** A perfectly predictable process.

Deterministic vs Random Processes

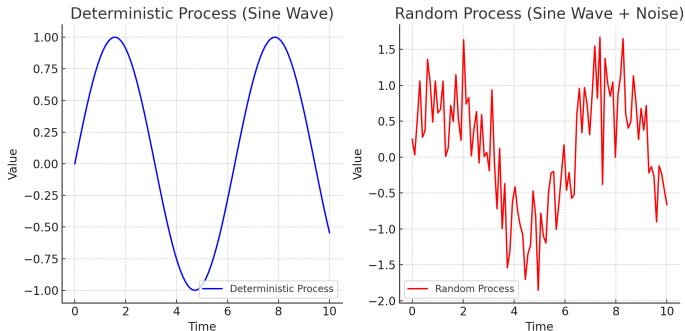


Figure 1: Deterministic and Stochastic Processes

Goal of Probability Theory and Random Processes

Central premise

To exploit the predictability of a random process to the maximum extent possible.

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With the backdrop of uncertainty, the simplest approach is to list all possible predictions and postulate the chances of each outcome.

Realizations of time-series

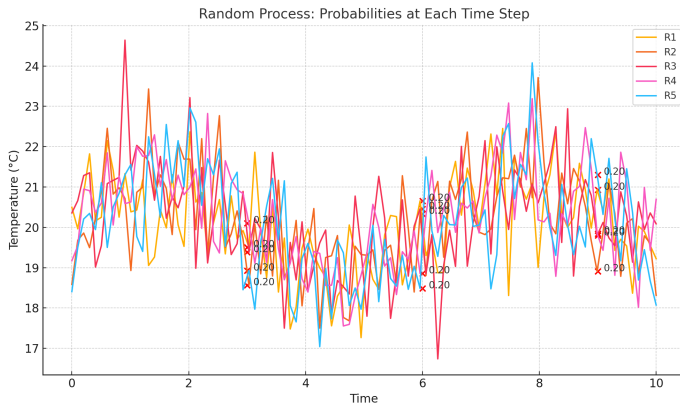


Figure 2: All possible realizations and their chances

Random Variables

Definition: A random variable is a function that assigns a numerical value to each outcome in a sample space.

- ▶ Random variables can be **discrete** or **continuous**.
- ▶ Discrete: Takes specific values (e.g., number of students).
- ▶ Continuous: Takes any value within a range (e.g., temperature).

Formal Definition of Random Variable

Definition: A random variable, X is a point function that maps from the sample space $\mathcal{S}$ to the real line $\mathbb{R}$, such that for each element $s \in \mathcal{S}$, there corresponds a unique real number.

- ▶ Example: A die roll is a random variable that maps each outcome $\{1, 2, 3, 4, 5, 6\}$ to the real numbers.

Introduction to Probability Theory

Probability Theory: A formalization of intuitive methods used to describe uncertainties.

- ▶ Probability of rolling a 6 on a single die roll.
- ▶ Estimating the chance of rainfall exceeding 100 mm in a given day.
- ▶ Predicting the likelihood of a stock price crossing a certain threshold within a week.

Estimates as Random Variables

Estimates: In probability theory, estimates also inherit the uncertainties of measurements.

Probability Distribution Functions

Definition: A probability distribution function (PDF) gives the probability that a random variable X takes a value within a specific range.

- ▶ The probability density function (PDF) for continuous variables:

$$f_X(x) \geq 0 \quad \text{and} \quad \int_{-\infty}^{\infty} f_X(x) dx = 1$$

Cumulative Distribution Function (CDF)

- ▶ The cumulative distribution function (CDF) for continuous variables:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

- ▶ The CDF for discrete variables:

$$F_X(x) = \sum_{t \leq x} P(X = t)$$

These distributions are either be theoretically known or experimentally determined.

Probability Dist. (Mass) Functions and Cum. Dist. (Mass) Functions

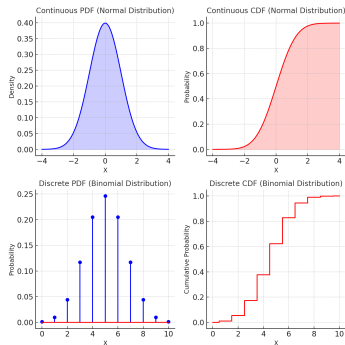


Figure 3: All possible realizations and their chances

Gaussian Function

Normal Distribution: A continuous probability distribution defined by its mean μ and standard deviation σ .

Normal Distribution: Probability Density Function

The probability density function (PDF) of the normal distribution is given by:

$$f(x|\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- ▶ μ (related to first parameter): Determines the location (center) of the distribution.
- ▶ σ (related to second parameter): Determines the spread or width.

Chi-Squared Density Function

The probability density function (pdf) of the chi-squared distribution with k degrees of freedom is:

$$f(x; k) = \frac{1}{2^{k/2} \Gamma\left(\frac{k}{2}\right)} x^{\frac{k}{2}-1} e^{-x/2}, \quad x > 0 \quad (1)$$

- ▶ $\Gamma(\cdot)$ is the gamma function.
- ▶ k is the degrees of freedom.

Usage of Chi-Squared Distribution

Relation to Gaussian Variables:

- ▶ If $Z_i \sim N(0, 1)$ are independent, then:

$$X = \sum_{i=1}^k Z_i^2 \sim \chi^2(k) \quad (2)$$

Applications:

- ▶ **Statistical Tests:** Goodness-of-fit, test of independence.
- ▶ **Confidence Intervals:** Estimating variance.
- ▶ **ANOVA:** Analysis of variance techniques.

Practical Aspects of PDF Estimation

- ▶ **Finding PDF apriori is non-trivial:** Many distributions do not have an obvious or known analytical form, requiring advanced techniques or assumptions.
- ▶ **Empirical determination of PDFs:** Estimating the PDF from data can be challenging, especially with limited data and the lack of efficient estimation methods (e.g., non-parametric methods such as kernel density estimation).
- ▶ **The multivariate challenge:** Estimating joint probability density functions for multivariate data is significantly more complex, as joint dependencies between variables must be captured.

- ▶ **Linear Random Processes:** In the case of linear random processes, it is not necessary to know the full PDF. Instead, it suffices to know the first two moments (mean and covariance).
- ▶ **First and Second Order Statistics:** For linear models, the estimates of model parameters, prediction errors, and bounds all rely on the first and second order statistical properties (mean and covariance) of the data's random characteristics.

Useful references / pre-requisites

- ➊ **Introduction to Statistical Hypothesis Testing** as contained in the video lectures of **NPTEL / MOOC** (<http://www.nptel.ac.in/courses/103106120/>)
- ➋ Basics of linear and matrix algebra.
- ➌ **Probability Distributions (Densities) and Moments** as contained in **Lectures 01A - 09B** of **NPTEL / MOOC** video lectures on **ATSA** (<http://www.nptel.ac.in/courses/103106123/>).
- ➍ **R Tutorials for Beginners**
 - ▶ <https://youtu.be/1mQIEdhPv6I>
 - ▶ <https://youtu.be/hvXr2MHIexQ>

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